

A portrait of Mireille El Gheche, a woman with dark, wavy hair and round glasses, wearing a black sleeveless top. She is leaning against a wooden post outdoors, with green foliage in the background. There are decorative circular overlays: a large grey one on the right and a smaller orange one on the left.

Mireille El Gheche
Lead Research Scientist
Softbank Robotics, Ex-Sony AI

The Free Spirit

How a woman who couldn't stand table tennis spent five years teaching a robot named after a samurai to beat elite players — and proved that different was exactly what the job needed.



"I'm very sporty. I love racquet sports: tennis, squash, padel, badminton, all of it."

In Zurich's Arterpark, Mireille El Gheche ties back her hair, takes her position at a table tennis table, and shows the photographer what five years of research looks like in her hands. Topspin, then backspin, precise and unhurried. "I got better," she says, laughing.

The irony amuses her more than anyone. "I'm very sporty. I love racquet sports: tennis, squash, padel, badminton, all of it. But the one sport I genuinely couldn't get into was, of all things, table tennis." For five years, that was exactly the game she was trying to make a robot master.

El Gheche, 39, grew up in Lebanon's northern mountains, one of five children, before the family settled in Tripoli, where she studied electrical engineering in a class split almost exactly 50/50 between

women and men. Second-oldest of the siblings, she was nonetheless the first to leave home, moving straight to Paris for a master's in radiocommunication systems.

"My parents were proud — it was a big moment for all of us," she says. It was her first time abroad. "It helped that I spoke the language," she says, "but it was completely different from what I had imagined. Everything so organised — the bus stops exactly where it's supposed to. In Lebanon you negotiate with the driver."

A PhD in optimization followed, then postdocs in Bordeaux and at EPFL, a path that read like a slow drift from anything resembling a table tennis table. Robotics, when it found her, arrived, in her own words, "almost unexpectedly." While interviewing with Sony AI in 2021, she was sent two papers to read, one on

computer vision and one on table tennis. Nobody told her what the project actually was; she worked it out herself, got it right, and joined as its first Senior Research Scientist, leading a group that would grow, eventually, to ten. It wasn't long before her influence outgrew it: she began reshaping how the whole project ran, well past her own corner of it. "I was given unusual freedom and trust from the start," she says, "not only to lead my own area, but also to influence how the wider project was organised."

At the outset, she was the only woman on the team — in a field still dominated by men, hardly a surprise. Her response was never to make herself smaller in it. "I'm not afraid to be in a room where I am different," she says. Her advice to other women follows the same logic: "If there are opportunities, take them." Don't let yourself think you're not completely ready for it.

What she was ready for turned out to be half a decade inside one narrow,

punishing problem. Every individual piece of the system worked, she has said; the difficulty was making them work together in real time, against a ball spinning too fast for a human eye to track. "Robotics is where intelligence has to deal with reality," she says. The robot, a six-jointed arm on a mobile base rather than a humanoid balancing on two legs, has milliseconds to read the ball, judge its path, decide how to respond, and move fast enough to return it — a skill honed over thousands of hours of simulated matches.

The team nicknamed the robot Musashi, after the fabled samurai said to have survived every duel he fought. His gift, famously, was adapting instantly to whatever opponent stood before him, never settling on one style. Musashi the robot, by contrast, was considerably less deadly. "He felt like a team member," she says. That sense of camaraderie was put to the test in the project's final weeks, as the work intensified in Tokyo, where she and her colleagues spent weeks in a

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tight loop of testing, fixing and iterating. "There was a lot of adrenaline," she says, "but it was also incredibly rewarding."

By the time the paper reached the cover of *Nature* this past spring, Musashi had beaten three of five elite players it faced, though it lost both matches against genuine professionals — a more clear-eyed tally than some headlines suggested. The elite player Rui Takenaka later exploited a blind spot: feed it a plain, spinless serve, and its own return comes back easy to punish — "the key reason," he said, "why I was able to win."

None of the work, El Gheche insists, was a solitary achievement. Much of it, in the end, was about bringing people together around shared problems — turning, as she puts it, "a difficult research problem" into "a real system." What she returns to more than the result, though, is how little the technical work and the human side of it were ever separate. "When you trust each other, it becomes much easier to challenge ideas, ask for help, and solve problems together," she says.

That same refusal, to simply accept set ways, turns up in

her Instagram bio too: "Free spirit, unfortunately for some." She borrowed the line from a biography of Princess Diana, whom she admires for having achieved so much, and for remaining unmistakably herself.

Meanwhile, the photographer has her popping the ball up off the bat, again and again, without ever looking down — a skill Musashi learned early on too.

On where robotics and AI go from here, she is "optimistic, but also slightly terrified." She pictures robots in hospitals, in care homes, woven as a genuine help into the small routines of everyday life. "I hope we can enable robots to help us," she says of the field she has just given five years to. Zurich, meanwhile, has earned her outright affection: "a robotics Mecca," she says, "I have found my dream city" — where she hikes on weekends and loses, cheerfully, at board games with friends. "I can lose," she says. "I just try not to."

She sends one more ball skimming low over the net, backspin curling it out of reach, and admits, smiling, that the sport has grown on her after all.

Text: Andrea Bleicher
Photos: Henrik Nielson

About the organizations



Greater Zurich Area Ltd.

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ETH AI CENTER

ETH AI Center

The ETH AI Center is ETH Zurich's central hub for artificial intelligence. We bring together researchers of AI foundations, applications, and implications across all departments. We foster research excellence, industry innovation, and AI entrepreneurship to promote trustworthy, accessible, and inclusive AI systems. As an interface between science, industry, and society, we create an environment where pioneering AI solutions can emerge and thrive.

Since its inception in 2020, the ETH AI Center has become a driving force in shaping Greater Zurich's AI ecosystem. We have mentored aspiring founders through our Entrepreneurs in Residence program, seen our fellows take up leading roles in academia and industry worldwide, grown a network of more than 60 affiliated startups across fields from robotics and AI safety to medical technologies and generative AI, and launched initiatives such as AI House Davos, the Zurich AI Festival, the AI Challenge for Teenagers, ICAIN, AI Launchpad, and the Swiss AI Initiative—which most recently released a fully open and transparent large language model as national infrastructure.

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